

VME PRECAST WORKSPACE

User Documentation

Integrated Assembly Management & BBS Sync System

Version 1.1.0

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Classification: Internal Use / Engineering

1. Overview

VME Precast Workspace is a Revit-integrated plugin designed for precast concrete engineers and detailers. It centralizes assembly management, bar bending schedule (BBS) generation, and automatic Revit sheet synchronization within a single unified interface.

1.1 Key Capabilities

- Manage multiple precast assemblies from a single dropdown selector
- Generate and view Steel (BBS), Concrete, and Hooks & Embeds schedules
- Automatically sync data to Revit sheets without manual re-entry
- Export individual assembly PDFs or batch-export all assemblies in one click
- View material take-off summaries (bar diameter, total bar length, weight)
- Access rebar bending schedule with full dimensional breakdown

1.2 System Requirements

Component	Requirement
Revit Version	Autodesk Revit 2021 or later
Operating System	Windows 10 / Windows 11 (64-bit)

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RAM	Minimum 8 GB (16 GB recommended)
Plugin Version	v1.1.0
Net Framework	.NET Framework 4.8 or later

2. User Interface Reference

The plugin interface is organized into four primary zones: the top toolbar, the assembly selector, the tab navigation panel, and the data schedule area.

2.1 Top Toolbar

Button	Color	Function
Sync All Assemblies	Blue	Pushes the current BBS/schedule data for all assemblies to their respective linked Revit sheets automatically.
Export PDF	Red/Pink	Exports the currently selected assembly's schedule data as a formatted PDF file.
Batch Export All	Orange	Exports PDF reports for all assemblies in the project simultaneously.
Help (?)	Dark Header	Opens the plugin help documentation or support page.

2.2 Assembly Name Selector

The Assembly Name dropdown (e.g., PW-GF-41) lists all precast assemblies available in the current Revit project. Selecting an assembly loads its associated schedules and data across all tabs.

NOTE: Assemblies are populated from Revit assembly definitions. If an assembly is missing, verify it has been created and named correctly in the Revit model.

2.3 Tab Navigation

Tab	Contents
Steel (BBS)	Bar bending schedule including summary material take-off and detailed rebar schedule with shape codes and dimensions.
Concrete	Concrete volume and mix details for the selected assembly.
Hooks & Embeds	Schedule of embedded hardware, inserts, and hooks specific to the assembly.

3. Steel (BBS) Tab — Detailed Reference

3.1 Summary Schedule (Material Take-off)

The summary table at the top of the Steel (BBS) tab aggregates rebar by diameter for the entire assembly:

Column	Description	Example Value
Bar Diameter	Nominal diameter of the rebar bar.	8 mm, 12 mm, 16 mm
Total Bar Length	Sum of all bar lengths for that diameter in the assembly.	242800 mm
Weight (kg)	Calculated weight based on length and bar density.	95.92 kg

3.2 Rebar Bending Schedule (BBS)

The detailed rebar bending schedule lists every individual bar or bar group in the assembly. The columns are defined as follows:

Column	Label	Description
Mark	PW-GF-41	Unique assembly mark identifier linking the bar to its parent assembly.
Type	T16 DOWEL BA	Bar type classification (T = deformed bar; number = diameter in mm; DOWEL BA = bar type).
Size	16 mm	Nominal bar diameter.
Shape	00, 26	BS 8666 shape code defining the bending geometry. Shape 00 = straight; Shape 26 = cranked/bent.
A	Dimension A	First leg length of the bar (mm).
B	Dimension B	Second leg length (mm); 0 mm for straight bars.
C	Dimension C	Third leg dimension (mm).
D	Dimension D	Fourth leg dimension (mm).
E	Dimension E	Fifth leg dimension (mm).
Bar Len	Bar Length	Total cut length of a single bar before bending (mm).
Total Len	Total Length	Bar Len x Qty — total material length required (mm).
Qty	Quantity	Number of identical bars required.
Weight	Weight (kg)	Total weight for that bar group (highlighted in blue).

4. Revit Sheet Synchronization

The Sync All Assemblies function automatically updates linked Revit sheets with the latest schedule data from the plugin. The status bar at the bottom of the window confirms the active sheet:

Active Revit Sheet set to: A-102 (Assembly Sheet – PW-GF-41)

4.1 How Sync Works

- Each assembly is linked to a specific Revit sheet (e.g., Sheet A-102 for PW-GF-41).
- Clicking Sync All Assemblies writes current schedule data to all linked sheets in the project.
- No manual placement of schedule views is required — the plugin handles placement automatically.
- The label next to the Assembly Name dropdown confirms: "Updates Revit Sheets automatically".

| NOTE: *Always save the Revit file after syncing to preserve the updated sheet data.*

5. Export Functions

5.1 Export PDF (Single Assembly)

- Select the desired assembly from the Assembly Name dropdown.
- Click Export PDF.
- Choose the destination folder in the save dialog.
- The exported PDF will contain the BBS, Concrete, and Hooks & Embeds schedules for the selected assembly.

5.2 Batch Export All

- Click Batch Export All to export PDFs for every assembly in the project in one operation.
- A destination folder is selected once; all PDFs are saved there with assembly names as filenames.
- Useful for issuing full project documentation packages.

| NOTE: *Batch export times depend on project size. Do not close Revit or the plugin during export.*

6. Status Bar

The bottom status bar provides real-time context:

Element	Description
Active Revit Sheet set to: A-102	Shows the Revit sheet currently linked to the selected assembly.
VME Precast Workspace v1.1.0	Displays the installed plugin version (bottom right).

7. Troubleshooting

Issue	Resolution
Assembly not in dropdown	Confirm the assembly has been created in Revit with a valid assembly name. Restart the plugin.
BBS data not showing	Ensure rebars are properly associated with the assembly in Revit. Check rebar host element.
Sync not updating sheets	Verify the sheet number linkage is configured. Check that the Revit sheet exists in the project.
PDF export fails	Ensure write permissions on the destination folder. Check available disk space.
Weight values appear zero	Bar density parameters may be missing. Confirm material parameters are set in Revit rebar families.
Plugin does not open	Confirm Revit version compatibility (2021+). Re-install the plugin if needed.

8. Glossary

Term	Definition
BBS	Bar Bending Schedule – a document listing all rebar with dimensions, shape codes, and quantities.
Assembly	A grouped collection of Revit elements representing one precast unit.
Shape Code	A standardized code (per BS 8666) describing the bend geometry of a rebar.
Sync	The process of writing schedule data from the plugin back to linked Revit sheets automatically.
Dowel Bar	A rebar type used to connect adjacent precast panels or foundations.
Material Take-off	A summary of all materials required, aggregated by type/size.